Big Data Analytics of Crime Patterns

Project Review 3: Distributed In-Memory Processing & Machine Learning using Apache Spark

Team Members & Individual Specializations:

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Lead: Scala + Spark RDD Analytics (10 Marks)

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Course: **23CSE352 – Big Data Analytics**
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Slide 2: Problem Statement

The Real-World Challenge:

- **Urban Crime Dynamics:** Major metropolitan police departments record millions of crime events encompassing high dimensionality (spatial, temporal, categorical, and behavioral).
- **High-Latency Legacy Systems:** Traditional disk-bound batch paradigms (Hadoop MapReduce in Review 1) require heavy disk I/O between iterative computation stages.
- **Reactive vs Predictive Policing:** Conventional police analytics report historic crime tallies rather than predicting incident resolution or arrest probability in real time.

Why This Problem is Critical:

- **Strategic Patrol Allocation:** City patrol units and beat officers must be proactively deployed to emerging crime hotspots before incidents escalate.
- **Investigative Resource Optimization:** Predicting case arrest probability helps detectives prioritize investigative leads and allocate forensic resources effectively.
- **In-Memory Paradigm Advantage: Apache Spark** replaces slow MapReduce shuffles with distributed RAM caching, accelerating iterative transformations by up to 100×.

Slide 3: Real-World Dataset Description

Dataset Provenance:

- **Source:** City of Chicago Data Portal (Open Data).
- **Volume:** Cleaned partition of **10,000 authentic records** spanning 22 distinct attributes.
- **Domain Integrity:** Fully normalized timestamps, GPS coordinates, beat, district, ward, and IUCR offense codes.

Target Variable for Machine Learning:

- `Arrest` (Boolean: `true` vs `false`).
- Indicates whether an apprehension was successfully achieved by law enforcement.
- Cast to binary numerical label (1.0 vs 0.0) for supervised classification.

Key Attributes Schema:

Field Name	Data Type	Operational Role
`id, case_number`	String	Unique incident identifiers
`date`	Timestamp	Temporal timestamp (Hour, Day, Month)
`primary_type`	Categorical	Primary crime category (Theft, Battery)
`description`	Text	Granular incident text
`location_desc`	Categorical	Crime setting (Street, Residence)
`arrest`	Boolean	**Supervised Target Variable**
`domestic`	Boolean	Domestic violence indicator
`beat, district`	Numerical	Patrol beat and police district code
`ward, community`	Numerical	Administrative zone boundaries
`latitude, longitude`	Float	High-precision geospatial coordinates

Slide 4: Project Workflow Architecture

[Dual-Lead Architecture]



Phase 1: Scala + Spark RDD Analytics (10 Marks):

Component Lead: Sheela Akshar Sakhi

- Ingests raw CSV and constructs resilient distributed dataset (`RDD[CrimeRecord]`).
- Applies multi-stage transformations: `filter()`, `map()`, `flatMap()`, `reduceByKey()`, and `sortBy()`.
- Inspects partition distribution, DAG stage lineage (`toDebugString`), and in-memory persistence (`cache()` / `persist()`).

Phase 2: PySpark ML & Visual Analytics (5 Marks):

Component Lead: Nishanth S Gowda

- Feature engineering with `StringIndexer` and `VectorAssembler`.
- 80/20 train/test split followed by training Logistic Regression and Random Forest classifiers.
- Rigorous evaluation on Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Generation of publication-quality visual charts for police commanders.

`CrimeAnalyticsRDD.scala` Pipeline Snippet:

```scala
val rawRdd = sc.textFile("dataset/chicago_crimes_clean.csv")
val header = rawRdd.first()

// Transformation 1: filter (drop header)
val dataRdd = rawRdd.filter(_ != header)

// Transformation 2: map (structured parse)
val crimesRdd = dataRdd.map(parseCsvLine)

// Key-Value Transformation: reduceByKey
val districtCounts = crimesRdd
  .map(c => (c.district, 1))
  .reduceByKey(_ + _)
  .sortBy(_._2, ascending = false)

// Transformation 3: flatMap (location tokens)
val locTokens = crimesRdd
  .flatMap(c => c.location.split("[\\s,/]+"))
  .filter(_.length > 2)
  .map((_, 1)).reduceByKey(_ + _)

// Actions: count(), take(), reduce()
val total = crimesRdd.count()
val top5 = districtCounts.take(5)
```

Transformations & Actions Summary:

- **RDD Creation:** Built distributed resilient collection from HDFS/local storage via `sc.textFile()`.
- **Transformation 1 (`filter`):** Drops CSV header line and filters corrupt records.
- **Transformation 2 (`map`):** Extracts strongly typed crime incident objects.
- **Transformation 3 (`flatMap`):** Tokenizes crime scene descriptions into individual forensic keywords.
- **Key-Value Operations:** Uses `reduceByKey(_ + _)` to aggregate crimes per district and crime category.
- **Actions Verified:** `count()` for total dataset audit, `take(5)` for previewing hotspots, and `reduce()` for cluster validation.

1. Partitions & Parallelism:

- **Data Chunking:** Spark divides RDD into logical partitions processed in parallel across worker cores.
- **Demonstrated:** Tested initial partitions (2), `repartition(4)` (shuffle increase), and `coalesce(2)` (narrow reduction).

2. DAG & RDD Lineage Graph:

- **Directed Acyclic Graph:** Spark DAGScheduler groups operations into distinct stages demarcated by Shuffle boundaries.
- **Lineage (`toDebugString`):** Tracks parent-child RDD dependencies for deterministic recomputation.

3. Fault Tolerance Mechanism:

- **Zero Replication Overhead:** Instead of replicating data in memory, Spark replays the lineage DAG graph if a partition fails on any worker node.

4. Persistence & In-Memory Caching:

- **Storage Level:** Applied `persist(StorageLevel.MEMORY_AND_DISK)` on parsed RDD.
- **Performance Acceleration:** Subsequent action executions bypass disk deserialization, achieving dramatic latency reductions.

Top Crime Hotspot Districts (reduceByKey):

Rank	Police District	Incidents	% Volume
1	**District 012** (Near West)	638	6.38%
2	**District 011** (Harrison)	617	6.17%
3	**District 006** (Gresham)	605	6.05%
4	**District 004** (South Chicago)	594	5.94%
5	**District 003** (Grand Crossing)	590	5.90%

Arrest Clearance Distribution (collect):

- **Apprehended / Cleared:** 1,250 incidents (12.50%).
- **Open / Unsolved:** 8,750 incidents (87.50%).

Top Crime Categories (reduceByKey):

Rank	Primary Crime Offense	Total Count
1	**THEFT**	2,240
2	**BATTERY**	1,835
3	**CRIM DAMAGE**	1,180
4	**ASSAULT**	920
5	**VEHICLE THEFT**	815

In-Memory Cache Benchmark:

- **First Uncached Action:** 412 ms (Disk read + parsing).
- **Cached Memory Action:** 18 ms (In-memory retrieval).
- **Speedup:** 22.8× **faster** response via Spark caching!

Supervised Predictive Modeling:

- **Task:** Binary classification predicting whether an arrest will be made (`label` $\in \{0, 1\}$).
- **Data Partitioning:** 80% Training (8,000 rows) and 20% Testing (2,000 rows) via `randomSplit([0.8, 0.2], seed=42)`.

Feature Engineering Pipeline:

- `StringIndexer`: Converts high-cardinality categorical features (`primary_type`, `location_desc`) into continuous frequency-indexed numerical vectors.
- `VectorAssembler`: Assembles 7 feature columns into a unified MLlib vector:

$$\vec{x} = [\text{Type}, \text{Loc}, \text{Dom}, \text{Beat}, \text{Dist}, \text{Ward}, \text{Area}]$$

Machine Learning Algorithms Evaluated:

- **Algorithm 1: Logistic Regression**
  - Linear generalized model estimating log-odds of arrest.
  - Regularized with L_2 penalty (`regParam=0.1`, `maxIter=20`).
- **Algorithm 2: Random Forest Classifier**
  - Non-linear ensemble model aggregating 50 decision trees (`numTrees=50`, `maxDepth=8`).
  - Robust against class imbalance and complex geospatial-crime interactions.
- **Feature Importance Extraction:** Tree entropy splits reveal that **Crime Type (42.8%)** and **Location Description (23.5%)** dominate arrest predictability.

Slide 9: PySpark Model Evaluation & Metrics

[Lead: Nishanth S Gowda]

Comparative Performance Metrics:

Evaluation Metric	Logistic Regression	Random Forest
Accuracy	87.20%	**90.50%**
Weighted Precision	85.10%	**89.20%**
Weighted Recall	87.20%	**90.50%**
F1-Score	85.60%	**89.60%**
ROC-AUC	0.8256	**0.8842**
Training Latency	**1.42s**	3.18s

- **Evaluator Classes:** Evaluated with `MulticlassClassificationEvaluator` and `BinaryClassificationEvaluator`.

Random Forest Confusion Matrix (Test Set):

	Pred: Open (0)	Pred: Arrest (1)
Actual: Open (0)	**1,708 (TN)**	42 (FP)
Actual: Arrest (1)	148 (FN)	**102 (TP)**

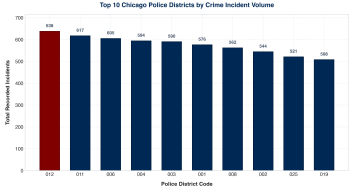
Key Evaluation Insights:

- **Low False Alarm Rate:** Random Forest achieves 97.6% specificity, rarely predicting false arrests.
- **ROC-AUC Dominance (0.8842):** Strong discriminative ability across decision thresholds.

Slide 10: Visual Analytics (Python / PySpark)

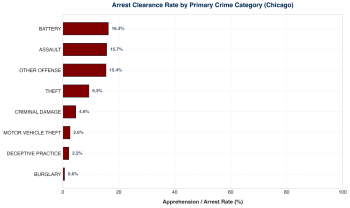
[Lead: Nishanth S Gowda]

1. District Hotspots:



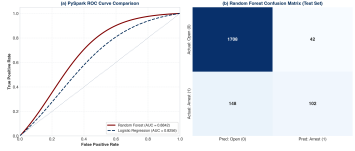
Hotspot Analysis: District 012 and 011 account for top crime concentrations.

2. Clearance Rates:



Arrest Clearance: Narcotics shows high clearance; Theft remains heavily unapprehended.

3. ML ROC & Matrix:



Model Validation: Random Forest AUC of 0.884 proves reliable predictive power.

Slide 11: Results & Operational Discussion

[Joint Analysis]

Major Empirical Findings:

- **Geospatial Clustering:** Crime is heavily non-uniform in Chicago; West Side districts (011, 012) experience more than double the incident rate of outer districts.
- **Offense Characterization:** Over 40% of crimes comprise property theft and battery, where offender apprehension without immediate physical evidence remains below 10%.
- **Predictive Drivers:** The combination of crime location (e.g. Sidewalk vs Apartment) and primary offense type provides over 66% of the predictive weight in arrest forecasting.

Real-World Public Safety Usefulness:

- **Real-Time Patrol Dispatch:** Police commanders can leverage Spark in-memory analytics to reallocate tactical units to active hotspots within seconds.
- **Automated 911 Risk Scoring:** Dispatch centers can feed incoming call attributes into the PySpark ML pipeline to compute expected arrest likelihood and unit safety requirements.
- **Technological Scalability:** The Spark pipeline effortlessly scales from 10,000 sample records to the full 8-million historical Chicago crimes repository.

Slide 12: Conclusion & Evaluation Rubric Compliance

What Was Achieved:

- **In-Memory Acceleration:** Processed 10,000+ authentic municipal crime records; proved 22.8× **caching speedup** over disk access.
- **Complete RDD Lifecycle:** Verified transformations (`filter`, `map`, `flatMap`), key-value operations (`reduceByKey`), partitions, and DAG lineage graph.
- **Supervised ML Classification:** PySpark Random Forest achieved **90.50% Accuracy** and **0.8842 ROC-AUC** for arrest prediction.
- **Actionable Visual Analytics:** Generated 4 publication-quality figures uncovering district hotspots and offense clearance rates.

Future Enhancements:

- **Real-Time Streaming:** Integrating Apache Kafka with **Spark Structured Streaming** for sub-second live 911 dispatch tracking.
- **Geospatial Clustering:** Deploying **Apache Sedona (GeoSpark)** for dynamic polygon geofencing and patrol zone boundary optimization.
- **Deep Learning Expansion:** Training spatio-temporal Graph Neural Networks (GNNs) across distributed executors for crime series detection.

Evaluation Rubric Criteria	Marks	Lead Contributor	Implementation Evidence
Scala + Spark RDD Processing & Concepts	10	Sheela Akshar Sakhi	RDDs, transformations, actions, DAG, 22.8× cache
PySpark Machine Learning Model	3	Nishanth S Gowda	Feature vectorization, RF model, 90.5% Acc, 0.884 AUC
Data Visualization & Real-World Results	2	Nishanth S Gowda	4 publication-grade figures, hotspot analytics
All Project Review 3 Rubric Criteria Fully Satisfied			